

Topics to discuss

- Mid point Integration
- Error in Integration
- Numerical Problem

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Midpoint Rule :

The midpoint rule is a numerical method used to find the approximate value of a definite integral.

In this rule, we divide the given interval into parts and take the middle point of each part. Then we calculate the function value at that middle point and use it to approximate the area.

Midpoint formula for one interval :

step size, $h = \frac{x_1 - x_0}{n} = x_1 - x_0$ or $b - a$

$$\int_{x_0}^{x_1} f(x) dx = (x_1 - x_0) f\left(\frac{x_0 + x_1}{2}\right) \quad \text{OR}$$

$$\int_a^b f(x) dx = (b - a) f\left(\frac{a + b}{2}\right)$$

for general mid-point rule from x_0 to x_n , when the interval is divided into n equal parts, the formula is,

$$\int_{x_0}^{x_n} f(x) dx = h \left[f\left(\frac{x_0+x_1}{2}\right) + f\left(\frac{x_1+x_2}{2}\right) + f\left(\frac{x_2+x_3}{2}\right) + \dots + f\left(\frac{x_{n-1}+x_n}{2}\right) \right]$$

Where,

$$h = \frac{x_n - x_0}{n}$$

Numerical Problem :

Q:- find the approximate value of $\int_0^2 \frac{1}{1+x^2} dx$

Using mid-point integration rule.

Solution: Using mid-point rule with $n=4$

$$\text{Step length, } h = \frac{b-a}{n} = \frac{2-0}{4} = 0.5$$

$$\text{So, } x_0 = 0$$

$$x_1 = 0.5$$

$$x_2 = 1$$

$$x_3 = 1.5$$

$$x_4 = 2$$

The subintervals are,

$$[0, 0.5], [0.5, 1], [1, 1.5], [1.5, 2]$$

Now the mid-points are :

$$m_1 = \frac{0+0.5}{2} = 0.25$$

$$m_2 = \frac{0.5+1}{2} = 0.75$$

$$m_3 = \frac{1+1.5}{2} = 1.25$$

$$m_4 = \frac{1.5+2}{2} = 1.75$$

Now lets make the table for, $f(x) = \frac{1}{1+x^2}$

x_{mid}	0.25	0.75	1.25	1.75
$f(x)$	0.9411764	0.64	0.390243	0.246153

Applying composite midpoint rule,

$$\begin{aligned}\int_{x_0}^{x_n} f(x) dx &= h \left[f\left(\frac{x_0+x_1}{2}\right) + f\left(\frac{x_1+x_2}{2}\right) + f\left(\frac{x_2+x_3}{2}\right) + f\left(\frac{x_3+x_4}{2}\right) \right] \\ &= 0.5 [0.9411764 + 0.64 + 0.390243 + 0.246153] \\ &= 0.5 \times 2.217573 \\ &= 1.108786500 \text{ ans.}\end{aligned}$$

Exact or True Value,

$$\begin{aligned}\int_0^2 \frac{1}{1+x^2} dx &= [\tan^{-1}x]_0^2 \\ &= \tan^{-1}(2) - \tan^{-1}(0) \\ &= 1.107148718\end{aligned}$$

$$\begin{aligned}\therefore \text{Absolute Error,} &= |\text{True value} - \text{approximate value}| \\ &= |1.107148718 - 1.10878662| \\ &= 0.001637902\end{aligned}$$

$$\begin{aligned}\therefore \text{Relative Error,} &= \frac{\text{Absolute Error}}{\text{True Value}} = \frac{0.001637902}{1.107148718} \\ &= 0.001479388\end{aligned}$$

$$\begin{aligned}\therefore \text{Percentage Error,} &= (\text{Relative Error} \times 100) \% \\ &= 0.147938 \%\end{aligned}$$

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